

Bishram Acharya

✉ [Email](#) 🌐 [Website](#) [LinkedIn](#) [GitHub](#) [Google Scholar](#)

Education

Bachelor of Engineering in Computer Engineering

Pulchowk Campus, Institute of Engineering, Tribhuvan University

April 2021 – Oct 2025

Lalitpur, Nepal

- Ranked **149th** in the nationwide engineering entrance exam out of 15,000+ applicants (**top 1.0%**)
- Graduated with First Division — **79.46%**
- Awarded Institute of Engineering Semester Scholarship seven times for top academic performance among peers
- Undergraduate Thesis : AI-Generated Image Detection via Frequency Domain Learning
- Relevant Coursework: Artificial Intelligence, Probability & Statistics, Numerical Methods, Engineering Math I, II & III, Applied Math, Applied Data Science, Computer Graphics, Simulation & Modeling, Data Structures & Algorithms, Theory of Computation

IELTS: 8.0 (Reading 9.0 , Listening 8.5, Writing 7.5, Speaking 7.5)

Publications

Automated Quality Assessment of Blind Sweep Obstetric Ultrasound for Improved

Feb 2026

Diagnosis

Prasiddha Bhandari, Nishant Luitel, Kanchan Poudel, **Bishram Acharya**, Angelina Ghimire,

Tyler Wellman, Kilian Koepsell, Pradeep Raj Regmi, Bishesh Khanal

Medical Imaging with Deep Learning (MIDL), 2026 [[Paper](#)]

- Systematically evaluated AI robustness to protocol deviations in sweep ultrasound videos (reversed direction, probe inversion, incomplete sweeps); developed automated quality-assessment models to detect perturbations and improve downstream task performance
- Simulated a clinical feedback loop in which flagged sweeps trigger re-acquisition, demonstrating measurable gains across three classification tasks: sweep-tag, fetal presentation, and placenta location
- *Primary contribution:* Ran scalable PyTorch Lightning and Hydra pipelines for systematic experimentation and reproducibility, while actively contributing to the preparation and writing of the manuscript

Research Experience

Research Assistant (full-time) | Supervisor: *Dr. Bishesh Khanal*

June 2025 – Present

NAAMII(TOGAI Lab) - Nepal

- **Core research direction:** Developing novel AI methods to analyse **fetal ultrasound videos** for automated diagnosis in **low-resource clinical settings**
- Contributing to industry R&D collaboration with **GE Healthcare** via NAAMII, working on solving real-world challenges of AI-assisted obstetric ultrasound
- Monitoring and analysing a **first-of-its-kind** blind sweep fetal video dataset collection, conducted in collaboration with **Institute of Medicine, Nepal**
- Research spans foundational models, domain adaptation across demographic datasets, label noise robustness, JEPA/world models, active learning, and multi-task learning
- Co-authored MIDL 2026 paper on automated quality assessment of blind sweep obstetric ultrasound

Research Intern | Supervisor: *Dr. Bishesh Khanal*

Sept 2024 – May 2025

NAAMII(TOGAI Lab) - Nepal

- Built an end-to-end pipeline for the **ISIC skin lesion classification challenge** including ensemble methods and test-time augmentation
- Co-developed a methodology combining **Active Learning and Vision-Language Segmentation Models (VLSMs)** for medical image annotation reduction; conducted systematic literature review and contributed to manuscript preparation

Industrial Experience

Freelance Developer

Nov 2023 – Aug 2024

Upwork (Remote)

- Built a Python application for automated news extraction and structured display from skynews.au
- Developed a data scraper for a real estate listings platform

Teaching Experience

Instructor

Aug 2025 – Oct 2025

AI-Assisted Disease Screening Bootcamp, NAAMII

- Designed and delivered a curriculum combining lectures, paper discussions, and a week-long project phase; trained participants to build AI-based tools for oral cancer screening

- Facilitated lab sessions on Graph Neural Networks (GNNs) created on top of lectures delivered by **Prof. Michael Bronstein and team**, Large Language Models (LLMs), and Vision-Language Models (VLMs)

Academic Projects

AI-Generated Image Detection | Final Year Thesis Project

- Implemented **FreqNet**, a frequency-aware deepfake detection model using convolutional layers in the frequency domain (using fast fourier transforms); investigated performance against CLIP - best model achieved **AUC 0.97**
- Found critical failure modes of deepfake detection models when fusing frequency domain and spatial domain features
- Evaluated performance across 10 datasets and 6 deepfake generation architectures
- Stack: Python, PyTorch, NumPy, OpenCV

Breast Cancer Classification - Hybrid Classifier | Solo Project

- Designed a custom hybrid architecture combining **ViT, LBP, and EfficientNet**
- Stack: Python, PyTorch, Scikit-learn, Pillow

Cross-Platform Fingerprint Matching System | Third Year Minor Project

- Trained a **Siamese network** (VGG-16 + DenseNet) for cross-platform fingerprint verification; achieved **89% accuracy**
- Stack: Python, TensorFlow, NumPy, Pillow

DevOps-Centric Image Classifier | Course Project

- Built a full-stack image classification web app (ResNet backbone) with a Node.js frontend and FastAPI backend; containerised and orchestrated via **Docker + Kubernetes**
- Stack: Python, PyTorch, FastAPI, Node.js, Docker, Kubernetes

Certifications

AI Course — Samsung Innovation Campus(SIC)

Samsung (Onsite in SIC, Nepal)

6 months

- ML (classification, clustering, anomaly detection), NLP, CNN/RNN/GANs, dimensionality reduction, language models

Deep Learning Specialization

Coursera — Andrew Ng / DeepLearning.AI

115 hrs

- Neural networks, CNNs, RNNs/LSTMs/GRUs, attention mechanisms, hyperparameter tuning, regularisation, transfer learning, sequence models, structuring ML projects

Generative AI with Large Language Models

AWS / DeepLearning.AI

16 hrs

- Transformer architecture, PEFT/LoRA, soft prompts, RLHF, multi-GPU training strategies, LLM evaluation benchmarks

Skills

ML / DL	Python, PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, OpenCV, Pillow
LLMs & GenAI	Transformer architectures, PEFT / LoRA, RLHF, prompt tuning, LLM evaluation
Engineering	FastAPI, Node.js, Docker, Kubernetes, Git, Unix/Linux, MySQL
Soft Skills	Clinical Collaboration, Technical Writing, Research Communication, Teaching

Research Interests

Self-supervised Learning, Vision-Language Models, World Models, Neural Cellular Automata, Domain Adaptation, Medical Imaging, Multimodal Representation Learning, Active Learning, Learning with Label Noise